ZIQI CUI

Google Scholar | LinKedIn | Github | Personal Website | Online Portfolio (2018-2023)

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- **Human Mobility, Human–Environment Interaction**
- **Spatially Explicit AI, Multi-Agent LLM, VLM, Explainable AI**
- **Geospatial Data Science, Urban Compute**

EDUCATION

Universitat Politècnica de Catalunya - Institute for Advanced Architecture of Catalonia, Spain **09/2022-09/2023**
M.Sc| Advanced Computation for Architecture & Design| GPA: 8.5/10 | Final Thesis: 9.3/10

Politecnico di Milano, Italy **09/2020-10/2023**
M.Sc| Architecture - Built environment - Interiors| GPA: 27.87/30 | Final Thesis: 110/110

Shanxi University, China **09/2015-06/2020**
B.Eng| Architecture| GPA: 82.56/100 | Final Thesis: 85/100

RESEARCH EXPERIENCE

Research Collaborator — The University of Milan, Dept. Computer Science

1. exp-EIA_D: SaaS, Experiential Simulation, Decision Support System, Digital Dashboard (Seed4Innovation Grant) **02/2024-06/2026**
- Led the development of a GNN-based perception prediction model (+3-8% accuracy vs. baseline) and integrated eye-tracking–derived attention features to enhance model explainability.
- Independently developed a 3D modeling plugin for predicting citizen feedback in urban design by C# and PythonNet.
- Contributed greatly to the development of a web-based urban decision-making platform by SQL data management and visualization.
- Led the development of a vision–language model based pipeline for fine-grained street-view image analysis, supporting downstream tasks.

Research Fellow — Politecnico di Milano, Dept. Architecture & Urban Studies

1. MED4PED: Mobility Experience Decision support system for pre-assessing Pededestrian walkability through on-site and off-site simulation (NextGenerationEU Grant) **02/2025-06/2026**
- Contributed greatly to the development of high-fidelity VR environment, experiment design, and experiment setup for walking behavior and environment perception study using wearable devices in both virtual and real-world settings.
- Contributed greatly to the multi-dimensional data collection, including EEG, ECG, eye-tracking, questionnaire, behavioral data and built environment data (urban form, 360° street view, climate data, etc) to build a comprehensive database.
- Led data processing (data synchronization , artifact removal, feature extraction) and quantitative analysis (PCA, t-SNE, correlation, regression, clustering, dimensional sentiment modeling) to extract interpretable perceptual features.
- Led the training of a multi-modal deep learning model to predict pedestrian visual attention in urban environments, based on large-scale fixation datasets (SALICON and CAT2000).

2. Multilayered Urban Sustainability Action: Urban Regeneration, Cities of the Future (NextGenerationEU Grant) **02/2024-06/2026**
- Led the development of an automated Retrieval-Augmented Generation (RAG)–LLM pipeline for systematic literature review in walkability research, identifying 101 variables and their effects, and constructing a dynamic knowledge graph to support the development of a novel walkability index.
- Independently completed quantification and visualization of 70+ urban morphological indicators across the Milan area via Python.

Research Intern — Politecnico di Milano, Dept. Architecture, Built Environment & Construction Engineering

1. Textiles Hub Interdepartmental Lab **04/2023-09/2023**
- Led the development of an adaptive prototype of a human-centered bending-active structure by C++ and Arduino, contributing significantly to the design and application research of smart building facades.

PUBILCATIONS

*+ first/co-first author; * corresponding author*

Conference Papers

1. Lou, S+*, & **Cui, Z.+**. (Under review). MoveMAS: Enhanced Human Mobility Prediction with Spatially-Aware via LLM-based Multi-Agent Systems. *ACM SIGIR 2026 Conference.* https://drive.google.com/file/d/1BDCbF0o-UQxajQIkchk_Zf8F1iis1DTh/view?usp=sharing (manuscript)
2. **Cui, Z.+***, & Lou, S+. (2025). Syncperception: A Real-Time Urban Perception Prediction Tool Based on Graph Neural Networks. *2025 Annual Modeling and Simulation Conference (ANNSIM)*, 1–13. https://ieeexplore.ieee.org/abstract/document/11118366
3. **Cui, Z.+***, & Lou, S+. (2025). How do the Spatial Layout of Street Elements and Geometric Features Affect Human Perception: A prediction model and explainable method based on Graph Neural Network. *ARCHITECTURAL INFORMATICS - Proceedings of the 30th CAADRIA Conference, Tokyo, 22-29 March 2025, Volume 4, Pp. 459–468*. https://doi.org/10.52842/conf.caadria.2025.4.459

Journals

1. **Cui, Z.+**, & Piga, B. E. A.* (Under review). How Urban Micro-Scale Built Environment Influences Perceived Walkability: A NLP-based Literature Review. *Building and Environment*.
2. Wei, B.+, **Cui, Z.+**, Wu, Q., Guo, S., Li, W., Wang, X., & Qiu, W*. (2025). Investigating the spatiotemporally heterogeneous effects of macro and micro built environment on sexual violence against women: A case study of Mumbai. *Humanities and Social Sciences Communications, 12*(1), 1–17. https://doi.org/10.1057/s41599-025-04838-4
3. **Cui, Z.+**, Zhang, S.+, Viscuso, S.*, & Zanelli, A. (2023). Weaving Octopus: An Assembly–Disassembly-Adaptable Customized Textile Hybrid Prototype. *Buildings*, *13*, 2413. https://doi.org/10.3390/buildings13102413

Book Chapters

1. **Cui, Z.+**, Piga, B. E. A.*, & Stancato, G. (2026). Impact of varying street view perspectives on urban perception: The case of Celoria Street in Milan. *Digital innovations in architecture, engineering and construction.* Springer. https://doi.org/10.1007/978-3-032-04711-3_24

CONFERENCE

Oral presentation

1. **Semantic Geospatial Retrieval of Urban Outdoor Spaces: Aligning Subjective Affective Needs with MLLM** — ***Ziqi Cui+****; Shangyu Lou+* — **AAG 2026**, San Francisco, California, USA *2026.03*
2. **Enhancing Human Mobility Prediction through an OSINT-Based LLM Multi-Agent System** — *Shangyu Lou+*; **Ziqi Cui+*** — **AAG 2026**, San Francisco, California, USA *2026.03*
3. **SyncPerception: A Real-time Urban Perception Prediction Tool Based on Graph Neural Network** — ***Ziqi Cui+****; Shangyu Lou+* — **SimAUD2025**, Madrid, Spain *2025.05*
4. **How Do the Geometric Features and Spatial Layout of Street Elements Affect Human Perception: A Prediction Model and Explainable Method Based on Graph Neural Network** — ***Ziqi Cui+****; Shangyu Lou+* — **CAADRIA2025**, The University of Tokyo, Tokyo, Japan *2025.03*
5. **Spatial-temporal Impact of Built Environment on Women's Safety in Mumbai** — *Baohua Wei+; **Ziqi Cui+**; Qilin Wu; Sitong Guo; Wenjing Li; Xinyue Wang; Waishan Qiu** — **ACSP2024 Annual Conference**, Seattle, USA *2024.11*
6. **In-motion Urban Ambiances: Assessing Walkability and Safety Perception Through Immersive VR Simulation** — *Barbara Ester Adele Piga+*, **Ziqi Cui**, Gabriele Stancato, Andrea Picardi, Simone Piersigilli, Lorenzo Mussone, Marco Boffi, Carlotta Colonna, Chiara Maninetti, Riccardo Barbieri, Luca Mainardi, Giandomenico Caruso* — **CitySenZ 2025**, Lisbon, Portugal *2025.07*
7. **Evaluating Urban Environment and Perception with Varying Street View Perspectives** — ***Ziqi Cui+****; Barbara Ester Adele Piga*; Gabriele Stancato* — **REAACH Conference 2024**, Online *2024.10*

Poster presentation

1. **"Flow along the corridor" —— A GAN based floor plan generator** — ***Ziqi Cui+****; Oana Taut; Xinyuan Zhang; David Andres Leon* — **Future of Construction (FoC) Conference 2024**, ETH, Zürich, Switzerland *2024.11*

ZIQI CUI

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TEACHING EXPERIENCE

Co-supervisor Master's Thesis

1. The impact of neighborhood built environment on the active travel behavior of older adults: Evidence from Milan — **09/2025-ongoing**
Student: Deng | Supervisor: Barbara Ester Adele Piga
- Supervise student in research framework development, literature review, and technical problem-solving.

2. Using Deep Learning Methods to Explore the Relationship Between Urban Elements at Intersections and Pedestrian Attention — **04/2024-04/2025**
Student: Li | Supervisor: Barbara Ester Adele Piga
- Supervised students in developing research frameworks and conducting literature reviews.
- Provided technical guidance in coding and machine-learning techniques.

3. Walkability in Urban Areas: Examining the Role of Built Environment and City Structure in Milan — **02/2024-07/2024**
Student: Mohammad Zare | Supervisor: Barbara Ester Adele Piga
- Guided student in conducting urban data mining and quantitative analyses.

Teaching Assistant

1. Master's Course: 055696 - URBAN SURVEY AND REPRESENTATION STUDIO — **09/2025-ongoing**
Professor: Eugenio Morello, Barbara Ester Adele Piga — **09/2024-01/2025**
- Delivered lecture: *Human-Centered Urban Computing: A Framework for Understanding Perception, Behavior, and Space*
- Assisted in project reviews and student supervision on urban quantitative analysis.

2. Master's Course: 061694 - ADVANCED METHODOLOGIES FOR HUMAN-CENTERED URBAN DESIGN — **02/2024-06/2024**
Professor: Barbara Ester Adele Piga, Mainardi Luca
- Supervised students in outdoor walking experiments with wearable sensors, focusing on data collection and processing.

WORK EXPERIENCE

Research Fellow

1. LABSIMURB Urban Simulation Lab — **02/2024-ongoing**
- Contributed greatly to data collection, quantitative analysis, method exploration and tools development for multiple EU-funded projects.

Intern

1. Textiles Hub Interdepartmental Lab — **04/2023-09/2023**
- Contributed greatly to the design and application of interactive intelligent building facade prototype.

2. Archiford Design ltd — **09/2022-10/2022**
- Led the design of a real-world experimental project using the discretization strategy.
- Independently developed structural details to support the implementation of the project.

3. Guangzhou City Construction and Design Company — **06/2019-12/2019**
- Independently responsible for 2D plans drawing and 3D modelling via AutoCAD and Rhino.
- Played a key role in the conceptual design for a commercial district planning project during the bidding phase.

SELECTED HONORS AND AWARDS

European Cooperation in Science & Technology (COST) Grant — **07/2025**
Awarded full travel and accommodation support to present at the CitySenZ Conference 2025 (oral presentation)

NextGenerationEU Research Scholarship Winner — **02/2024**
MED4PED: Mobility Experience Decision support system for pre-assessing Pedestrian walkability through on-site and off-site simulation

Graduated with Full Marks Honour from Politecnico di Milano — **10/2023**
Final Thesis: Weaving Octopus_An Assembly–Disassembly-Adaptable Customized Textile Hybrid Prototype.

Erasmus+ Scholarship Winner — **06/2022**
"Passion in Action" programme: Taking Nantes (France) as an example to explore pedestrians' multi-dimensional perception of the city.

EXHIBITIONS

Milano Design Week 2025: "ACTUATED KNIT// TEXTILEMORPHOSIS" — **04/2025**
Ziqi Cui, Siman Zhang, Salvatore Viscuso, Alessandra Zanelli

ACADEMIC WORKSHOP

Sponsored Participant

1. Mobile Brain/Body Imaging Workshop (TU Berlin) *Professor: Klaus Gramann* — **10/2025**
- Designed and conducted mobile EEG experiments exploring embodied cognition, attention, and perception in dynamic environments.
- Practically trained in EEG-based brain–body data acquisition, synchronization, and cleaning using the pipeline of EEGLAB, Lab Streaming Layer and MoBI Toolbox.

Selected Participant

1. DigitalFuture2023 Workshop (Tongji University&ETH) *Professor: Benjamin Dillenburger* — **06/2023**
- Led project and explored the automatic generation of high-rise building layouts by setting up a reward mechanism and training a Reinforcement Learning model.

2. Artificial Intelligence Urban Study Workshop (Polimi) — **07/2022-09/2022**
- Led project "Spatiotemporal Prediction Model on VAW Incidents in Mumbai in relation to Urban Environmental Factors" by proposing a gap, mining urban big data and establishing GWR, GTWR and baseline models

3. Smart City and Heritage Summer workshop (Polimi) — **07/2022-08/2022**
- Led project: Using Blender and Unity to revive architectural heritage through VR and AR in Cortona, Italy.

Erasmus+ Scholarship Winner

4. Experiencing Nantes, A psycho-architectural approach to urban study — **06/2022**
- Contributed greatly on data colloection and analysis to research: Taking Nantes (France) as an example to explore pedestrians' multi-dimensional perception of the city.

SKILL

Coding:
Python, MATLAB, C#, C++, JavaScript, Web.gl, PostgreSQL

Analysis&Simulation:
ArcGIS, QGIS, PTV Vissim, MATSim

3D Modeling:
Rhino, Grasshopper, Revit, Blender

Interaction:
Unity, UE5, Arduino, ARCore, AR Foundation

2D Drawing:
AutoCAD, PhotoShop, Ilustrator, InDesign

Langaue:
Chinese, English